Question 7: Prepare a monitoring, logging, and DevSecOps report describing the automation, observability setup, security findings, and observations with screenshots.

This report describes the complete Lab 3 workflow: automated configuration and provisioning, the observability setup, security findings and observations.

1. Objective

Build an automated, observable and secure infrastructure workflow around the Flask application: Ansible and Terraform for automation, Prometheus/Grafana for metrics, the ELK stack for centralized logging, and DevSecOps scanning plus feature flags, load and browser testing integrated into the process.

2. Tools and environment

- Ubuntu 24.04 LTS (WSL2), Docker Engine 29.9.2, Docker Compose v5.7.0
- Ansible core 2.21.5, Terraform v1.17.1 (docker provider)
- Prometheus v2.53, Grafana 11.1, Elasticsearch/Logstash/Kibana 8.14.1
- Trivy 0.76.1, pip-audit, bandit, Selenium 4 + Chrome/ChromeDriver 157, apache bench

3. Architecture

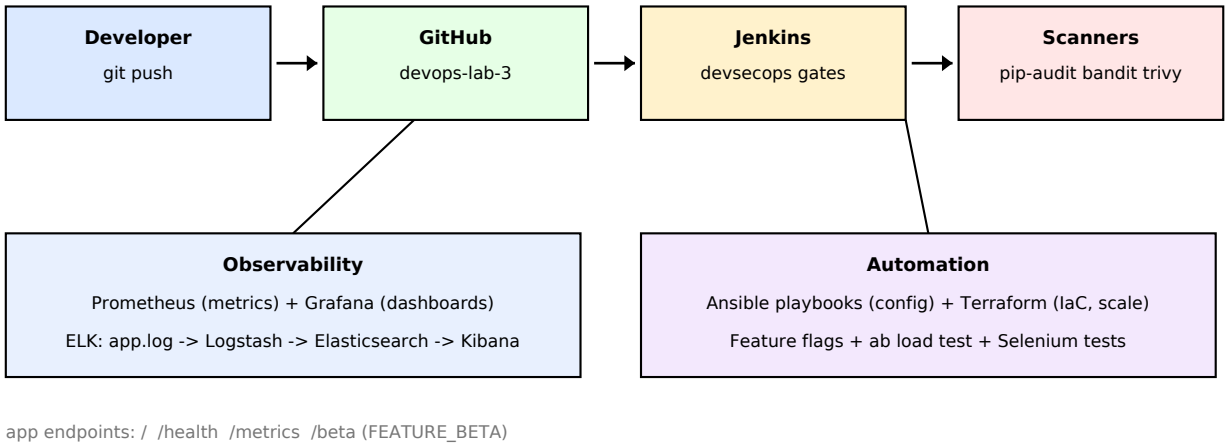


Fig: Lab 3 monitoring, logging and DevSecOps architecture

4. Automation results

Tool	What it automated	Evidence
Ansible	nginx install + app page on the managed node	playbook recap ok=6 changed=0 (2nd run)
Terraform	2 -> 4 nginx containers via one variable	apply outputs + terraform state list
Jenkins	security gates on every change (poll SCM)	builds 1-3 blocked, build 4 passed

5. Observability setup

- Metrics: prometheus_client counter on every response; 5s scrape; Grafana dashboard with req/s, totals and target uptime, provisioned from files.

- Logs: Flask file logging -> Logstash grok parse -> Elasticsearch index lab3-app-logs (765+ documents) -> Kibana Discover with structured level/message fields.
- Both observability stacks are single docker-compose files - reproducible on any machine.

6. Security findings

Scanner	Finding	Resolution
pip-audit	flask 3.1.0: PYSEC-2026-1377/2151; pytest 8.3.4: PYSEC-2026-1845	bumped to flask 3.1.3 / pytest 9.0.3 - audit clean
bandit	B108 hardcoded /tmp dir; B104 bind-all-interfaces	B108 fixed in code; B104 skipped with documented container justification
trivy	real HIGH/CRITICAL base-image CVEs (util-linux, perl-base) in the lab2 image	report-only; base image refresh queued as follow-up

7. Testing results

- Feature flag: same image, /beta 404 (off) vs 200 (on) - toggle without redeploy.
- Load test (ab, 2000 req, c=50): 323.74 req/s, 0 failed, p95 211 ms, p99 267 ms.
- Selenium (headless Chrome 157): /health asserted, enabled beta flag asserted, screenshot captured - 3/3 PASS.

8. Evidence

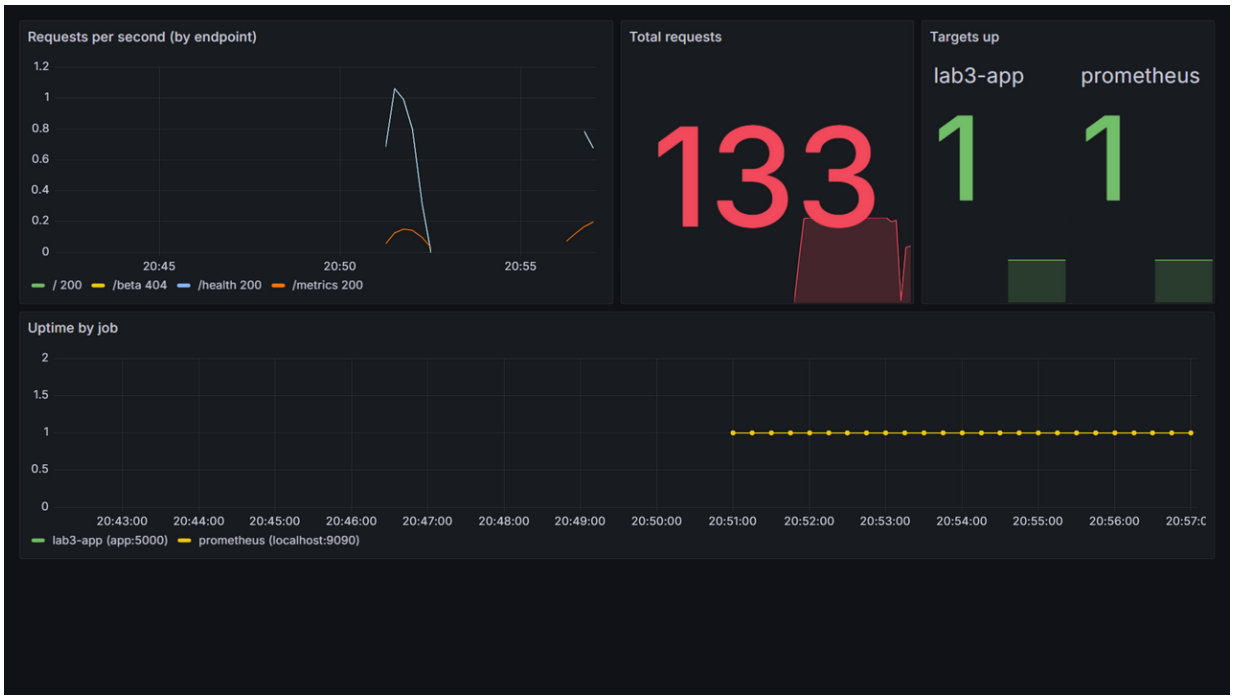


Fig: Grafana dashboard with live metrics

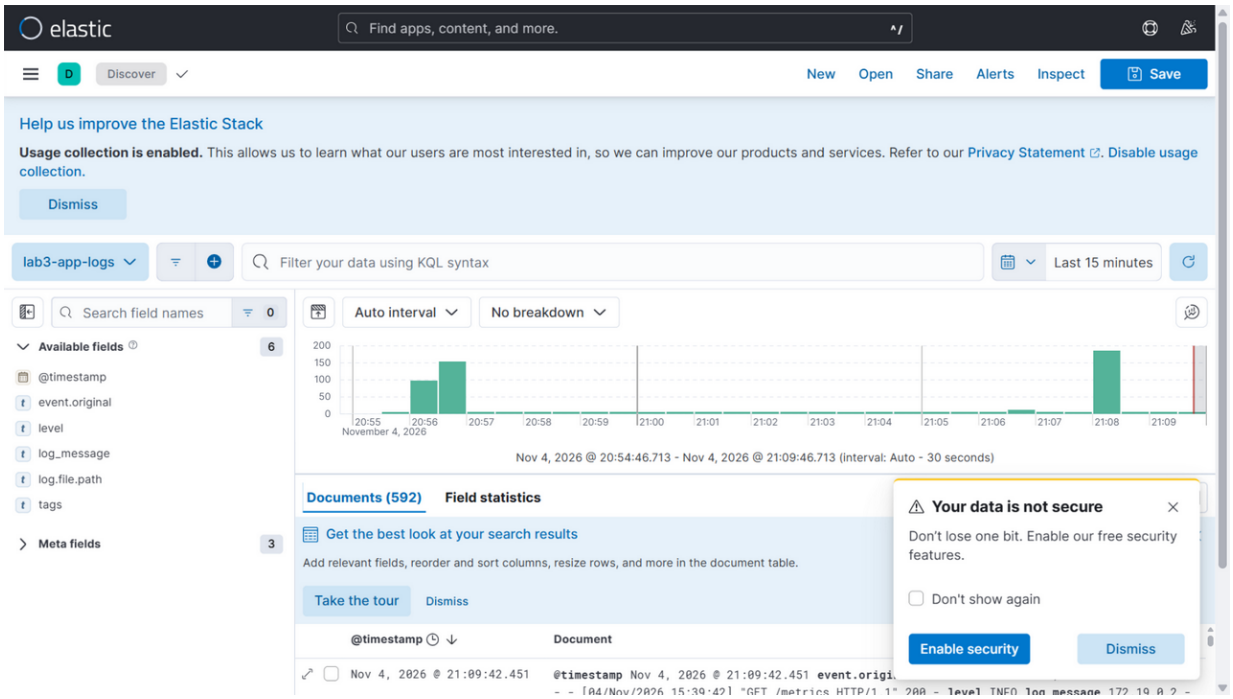


Fig: Kibana Discover with parsed logs

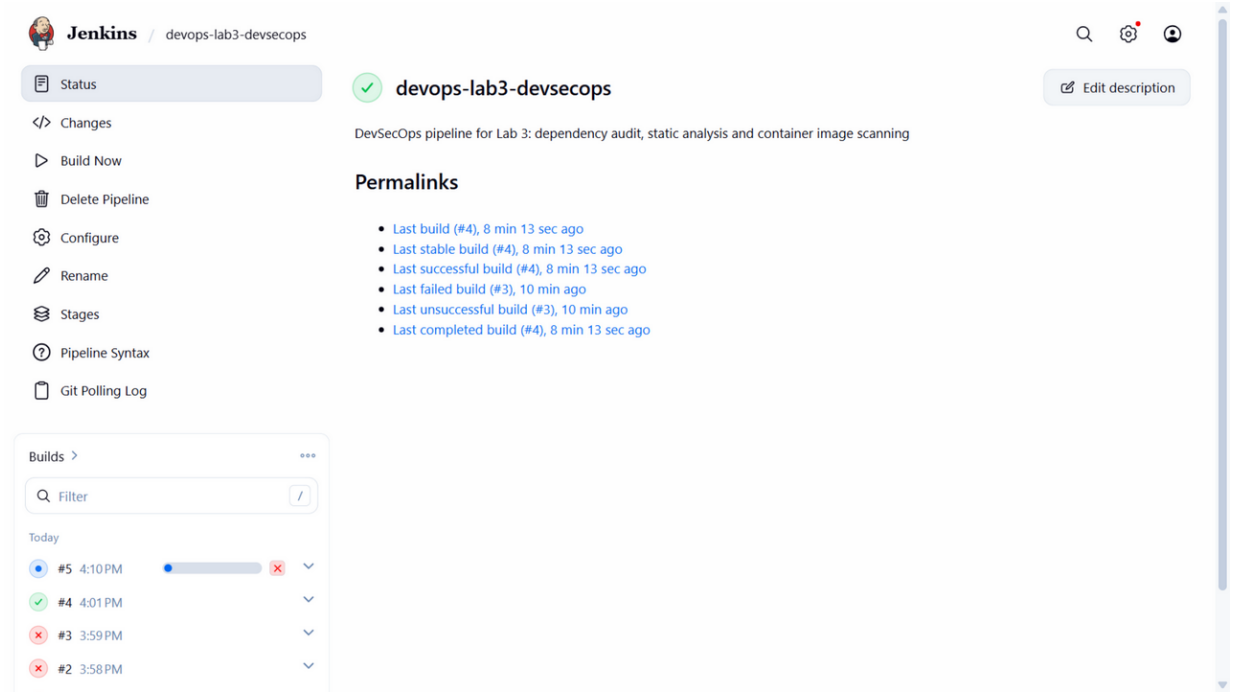


Fig: DevSecOps pipeline history

Pretty-print ☐

{"enabled":true,"feature":"beta","message":"Welcome to the beta feature"}

Fig: Selenium browser screenshot

9. Observations

- Ansible's docker connection plugin removed all SSH setup pain for the managed node.
- Terraform plan is the safety net: the scale change was reviewed as '2 to add, 0 to change' before applying.
- Provisioned Grafana (datasource + dashboard as files) means the dashboard exists on first boot - no clicks.
- Grok parsing is what makes logs searchable; raw lines alone would make Kibana useless.
- The security gate caught real advisories (flask 3.1.0) - proof that dependency scanning belongs in CI, not in an occasional manual audit.
- Bandit exceptions must be documented; B104 (bind all interfaces) is genuinely required inside containers.
- ELK needs host tuning in WSL2 (vm.max_map_count) and is memory hungry - sized down to 512m heap for the lab.

10. Conclusion

The workflow is now automated (Ansible, Terraform), observable (Prometheus/Grafana metrics, ELK centralized logs) and secure (pip-audit, bandit and trivy gates in the CI/CD pipeline, with two real blocks demonstrated). Feature flags, load testing and Selenium browser testing complete the modern DevOps practice loop. All evidence is committed in github.com/theabhinav0231/devops-lab-3.